Paper id: 250617

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VI) THEORY EXAMINATION 2024-25
AERODYNAMICS-II

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. 02 x 7 = 14

Q no.	Question	CO	Level
a.	Define critical Mach number and explain its significance.	1	K2
b.	Write the expression for the area-velocity relation for compressible flow.	1	K2
c.	State the difference between a normal shock and an oblique shock.	2	K2
d.	Mention two key features of a supersonic wind tunnel.	5	K2
e.	Define power loading in rotor blade theory and state its relevance.	6	K2
f.	Define Rayleigh flow.	1	K2
g.	What are the assumptions made in one-dimensional compressible flow theory?	1	K2

SECTION B

2. Attempt any *three* of the following: 07 x 3 = 21

a.	Sketch and explain the flow through a converging-diverging nozzle for subsonic, sonic, and supersonic conditions.	1	K2
b.	Derive the oblique shock relations for a supersonic flow over a wedge.	2	K3
c.	Explain the concept of compressibility correction and derive Prandtl-Glauert transformation.	3	K3
d.	Explain blockage and wall interference effects. How are they corrected?	4	K2
e.	Define solidity and its influence on propeller thrust and efficiency.	5	K2

SECTION C

3. Attempt any *one* part of the following: 07 x 1 = 07

a.	Derive the area-Mach number relation for compressible isentropic flow.	1	K3
b.	Explain the concept of Fanno flow. What are its characteristics?	1	K2

4. Attempt any *one* part of the following: 07 x 1 = 07

a.	If the entropy change caused by a normal shock in an airstream is 200 J/kg K, determine the Mach number ahead of the shock and the shock strength.	2	K3
b.	Define the Prandtl-Meyer function and describe its role in expansion waves.	2	K2

5. Attempt any *one* part of the following: 07 x 1 = 07

a.	Derive the linearized small perturbation potential equation for compressible flow.	3	K3
b.	What is center of pressure? How does it vary with Mach number in compressible flow?	3	K3

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	What are shock tunnels? Explain their working principle with a schematic diagram.	5	K2
b.	Explain the use of the following measuring devices in wind tunnel testing- (i) Pitot static tube (ii) Hot wire anemometers	5	K2

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	(iii) PIV technique (iv) Tuft survey		

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Derive an expression for induced power in hovering flight.Describe the control mechanisms used in helicopters: collective and cyclic pitch.	5	K2
b.	How does the ground effect influence the lift and induced drag of rotor blades?	6	K2

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